SRI CHANDRASEKHADRENDRA SARASWATHI VISWA MAHAVIDYALAYA

UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956

ENATHUR, KANCHIPURAM - 631 561 DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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REG NO :11249A389

CLASS : S4

SUBJECT CODE : BCSD244A12

SUBJECT NAME :DATA SCIENCE TOOLKIT LAB

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of work done by

Mr/Ms. UDAYAGIRIVEERAKOUSHIK with

Reg. No 11249A389 done by of II-B.E.(CSE) in –

DATA SCIENCE TOOLKIT LAB for Lab during the academic year 2025 2026.

Station : ENATHUR

DATE:-

Staff-in-charge **Head of the Department**

Submitted for the practical examination held on _____

Examine-1 **Examine -2**

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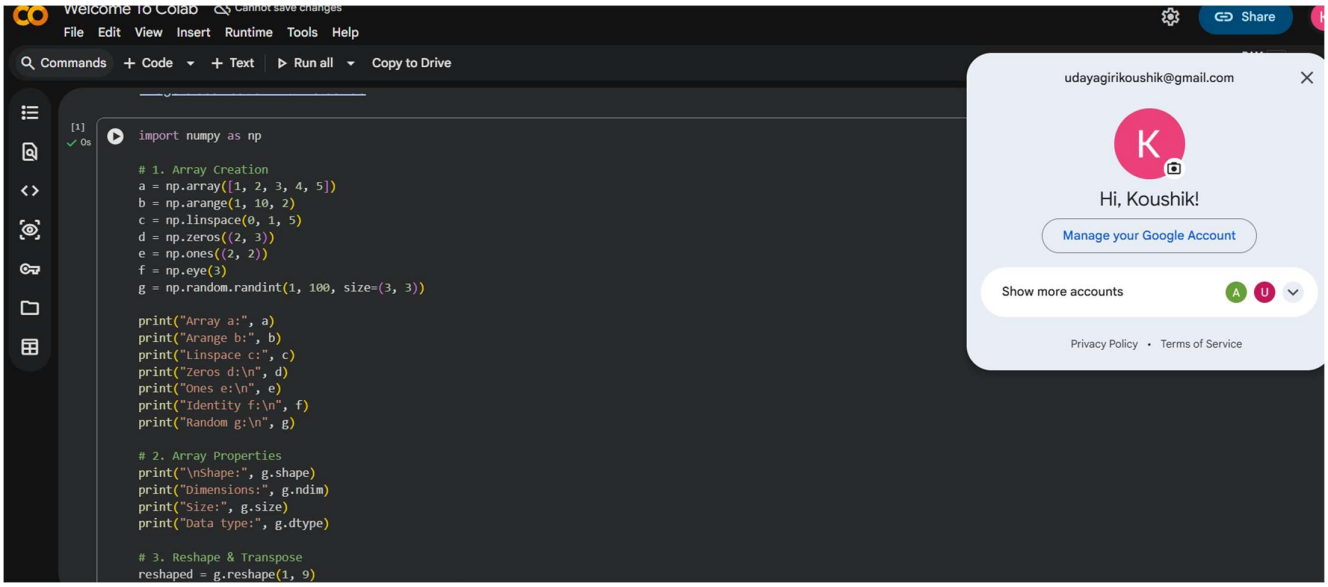
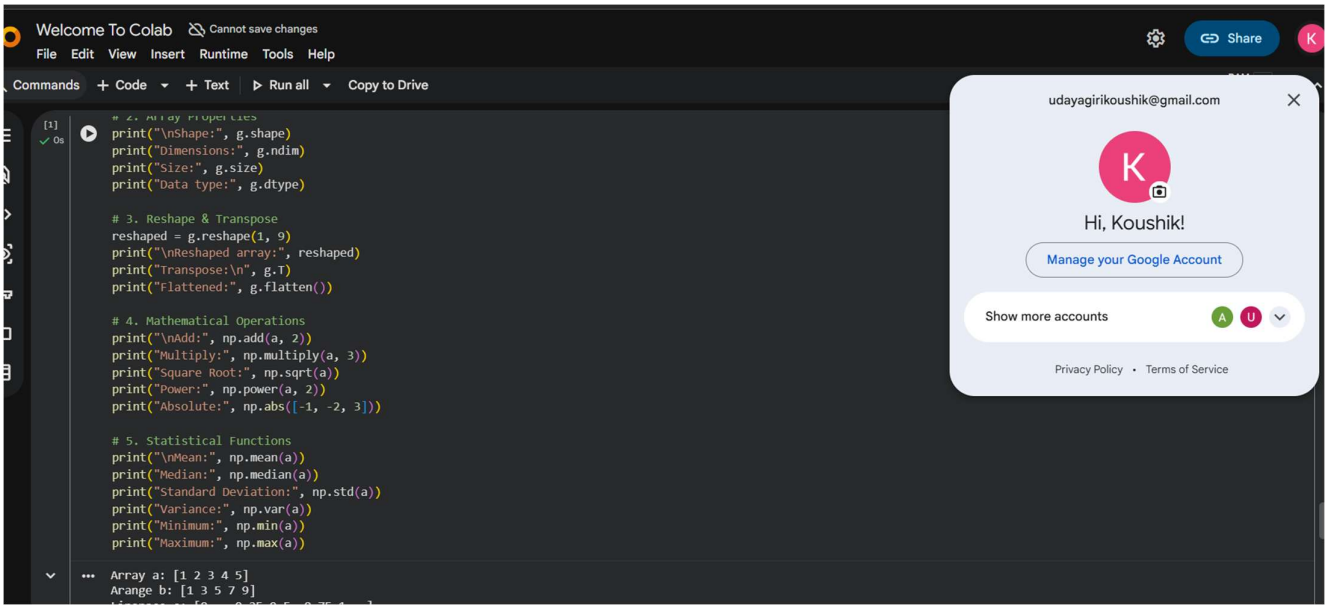
INTRODUCTION TO PYTHON LIBRARIES FOR DATA SCIENCE

Exp-01(1.1) Date-02-02-2026

Aim:

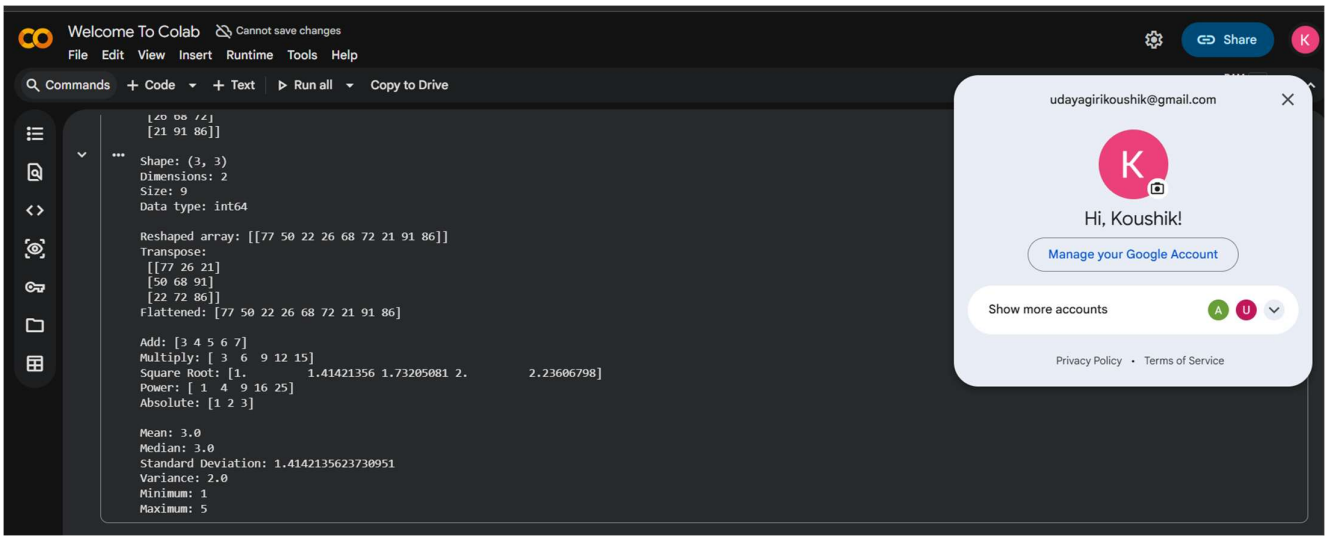
To learn basic Python data science libraries like NumPy and Pandas.





OUTPUT-

Screenshot showing output of PANDAS



Result:

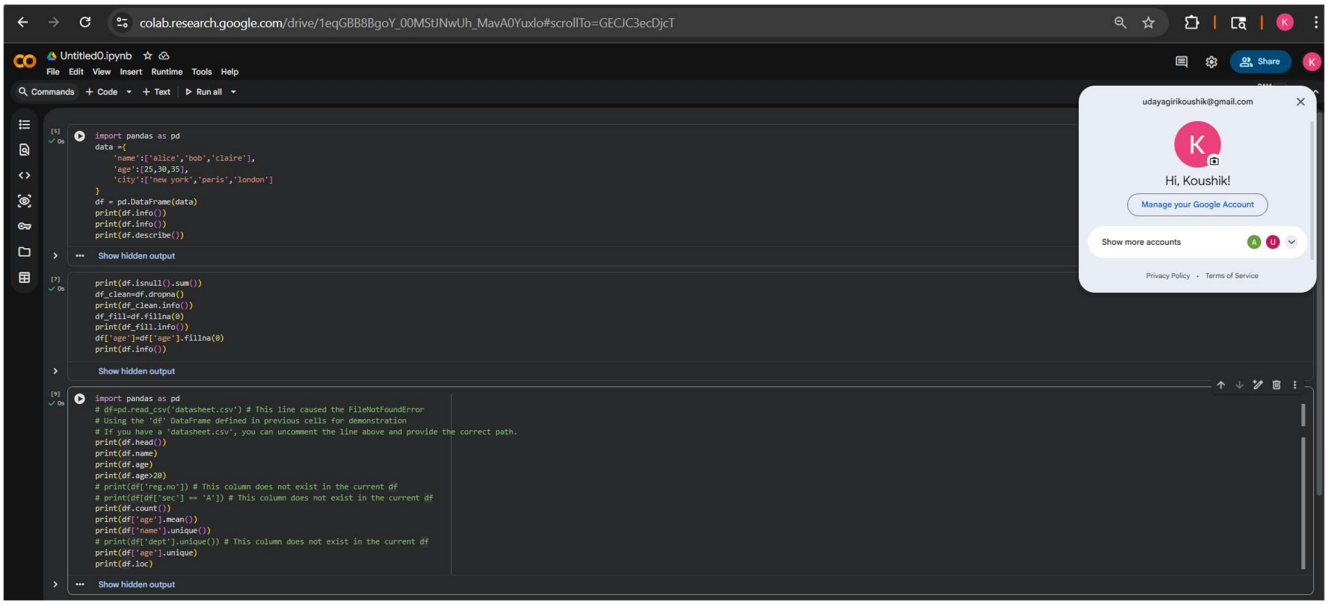
Thus, it is implemented *Introduction to Python Libraries for Data Science* successfully.

PANDAS

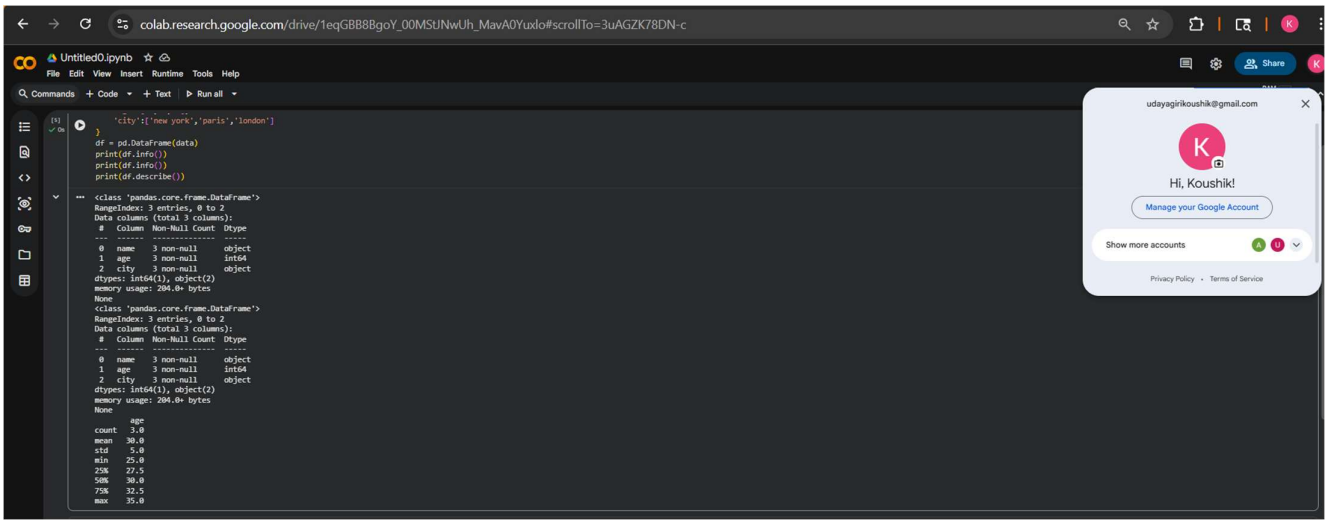
EXP-1.2 DATE-02-02-2026

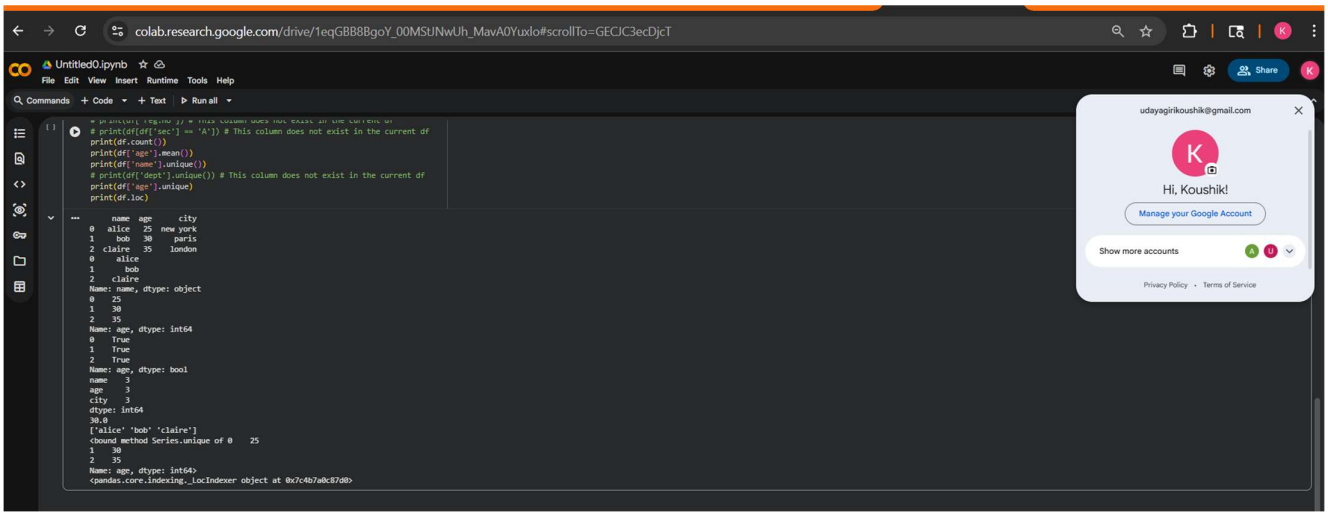
Aim:

To study the Pandas library for data handling and analysis.



OUTPUT-





Result:

Thus, it is implemented *Pandas* successfully.

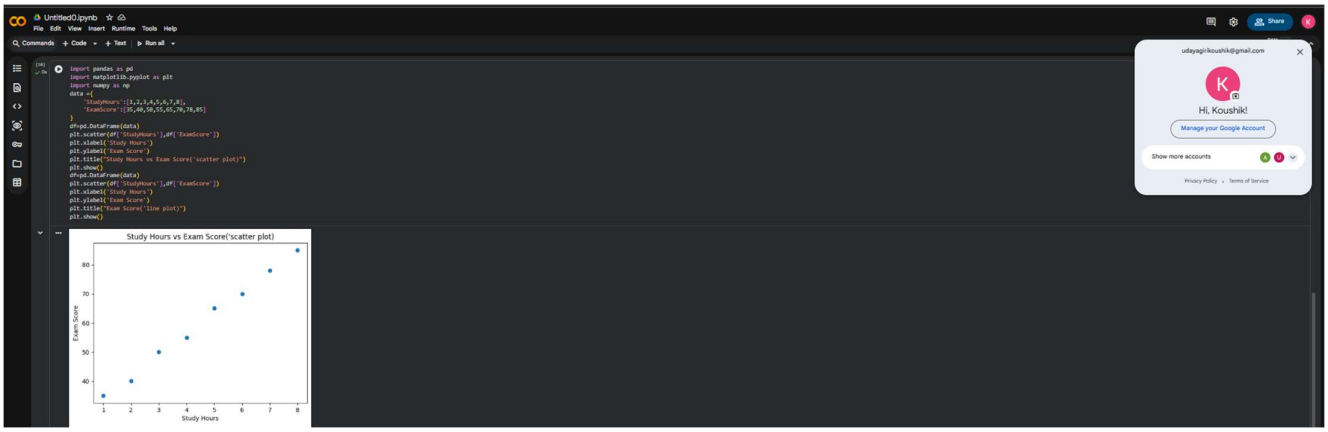
MATPLOT LIB

EXP-1.3 DATE-02-02-2026

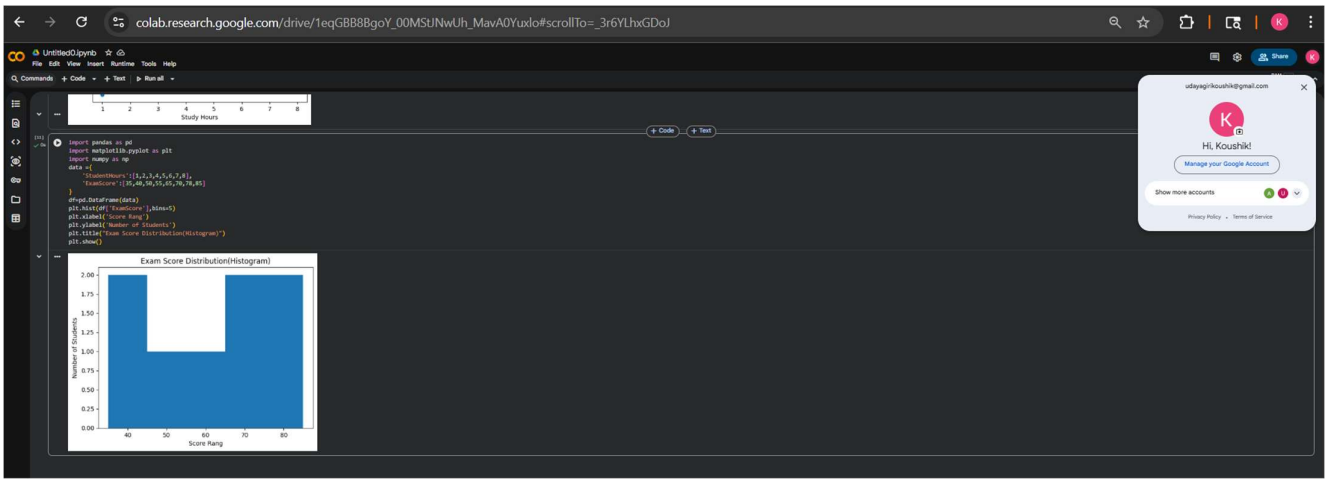
Aim:

To study the Matplotlib library for data visualization.

PROGRAM WITH OUTPUT-



PROGRAM WITH OUTPUT-



Result:

Thus, it is implemented *Matplotlib* successfully.

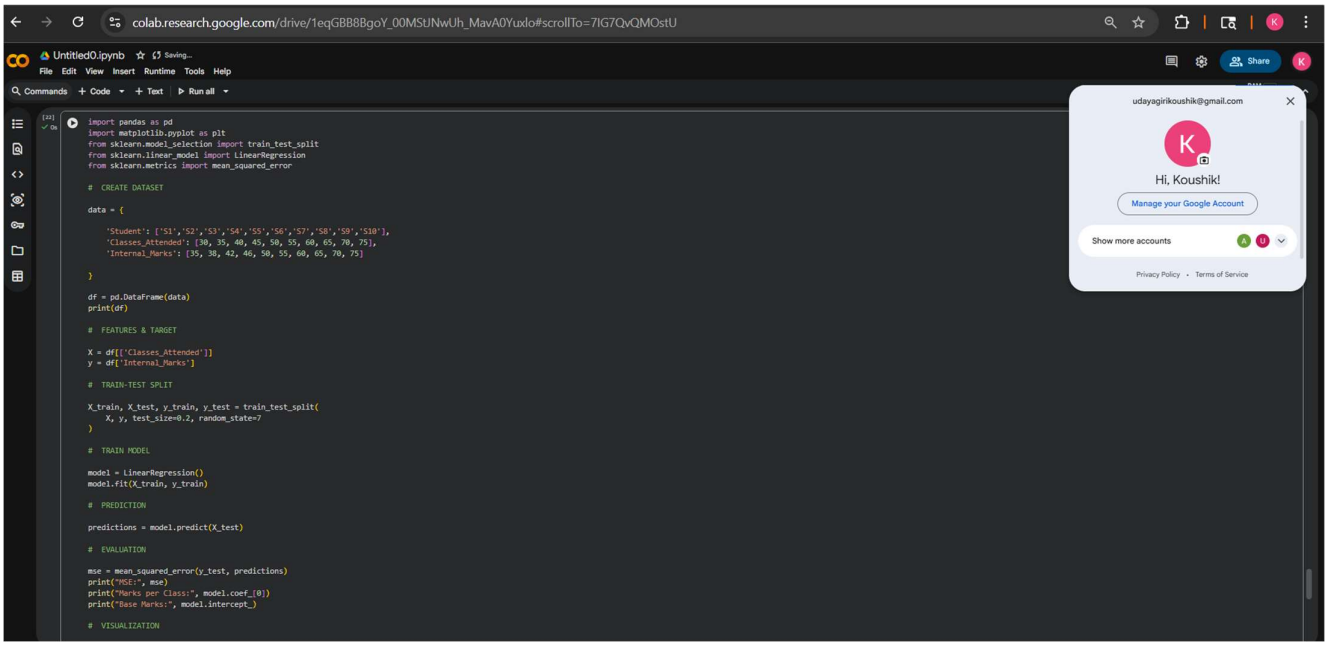
SCIKIT LEARN

EXP N-1.4 DATE-09-02-2026

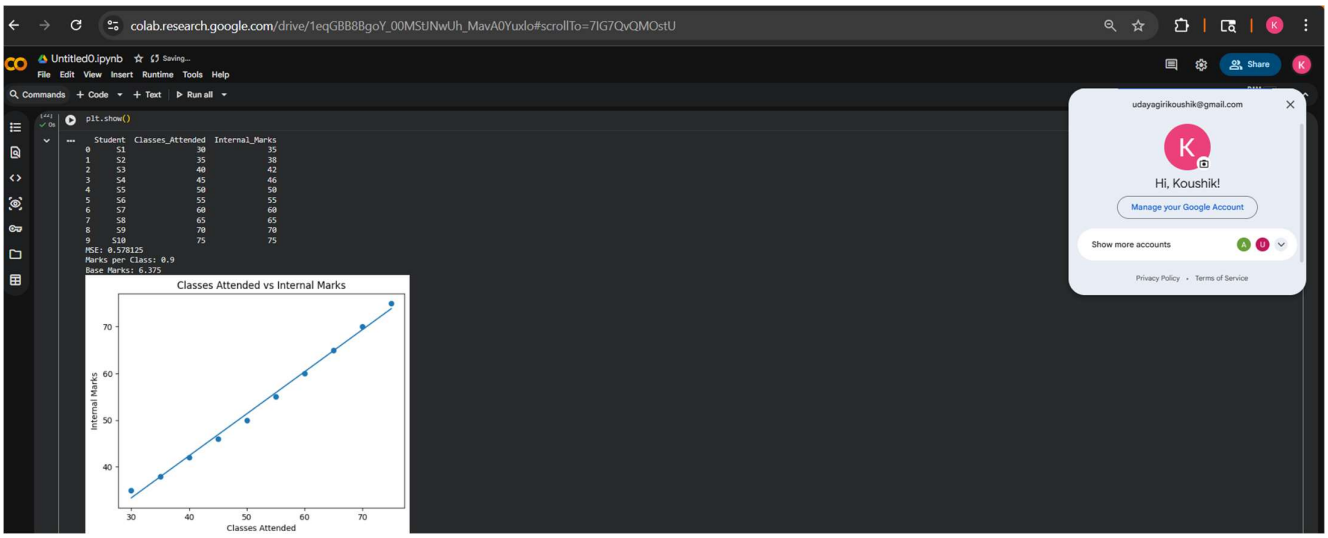
Aim:

To study the scikit-learn library for machine learning.

PROGRAM:



OUTPUT-



Result:

Thus, it is implemented *Scikit-learn* successfully.

PERFORM DATA EXPLORATION AND PREPROCESSING IN PYTHON

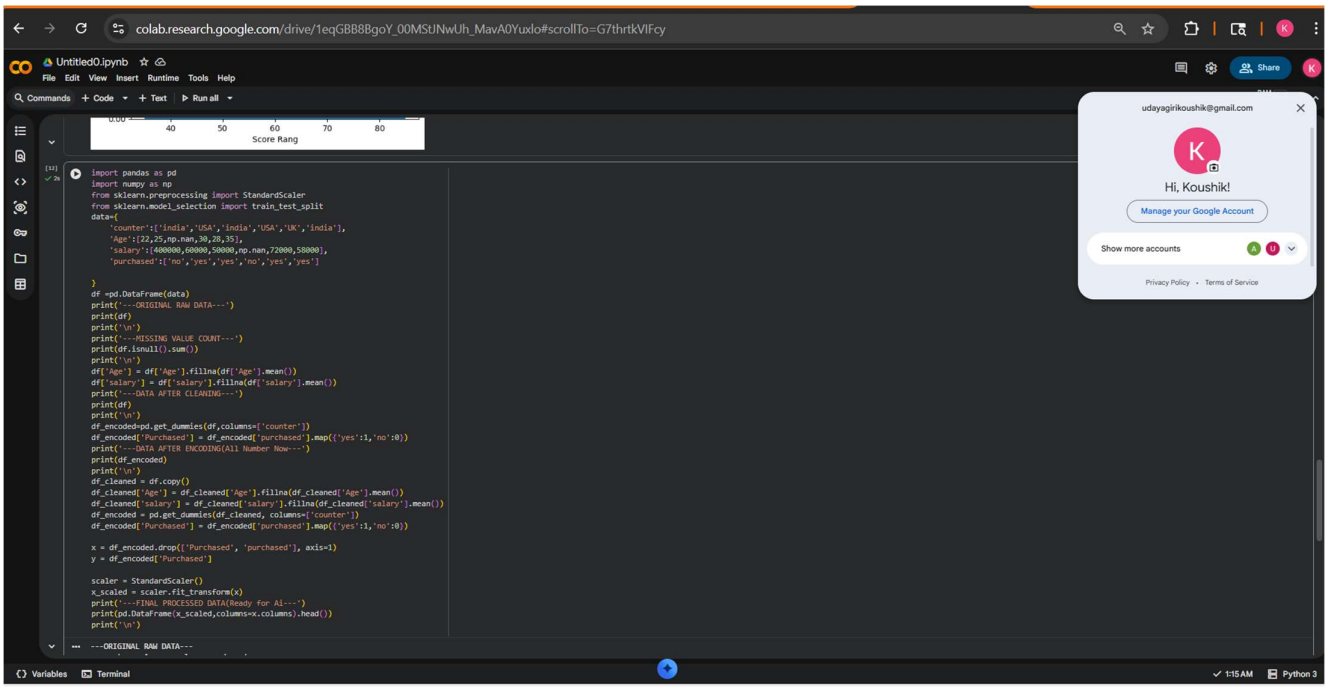
EXP-02

DATE-14-02-2026

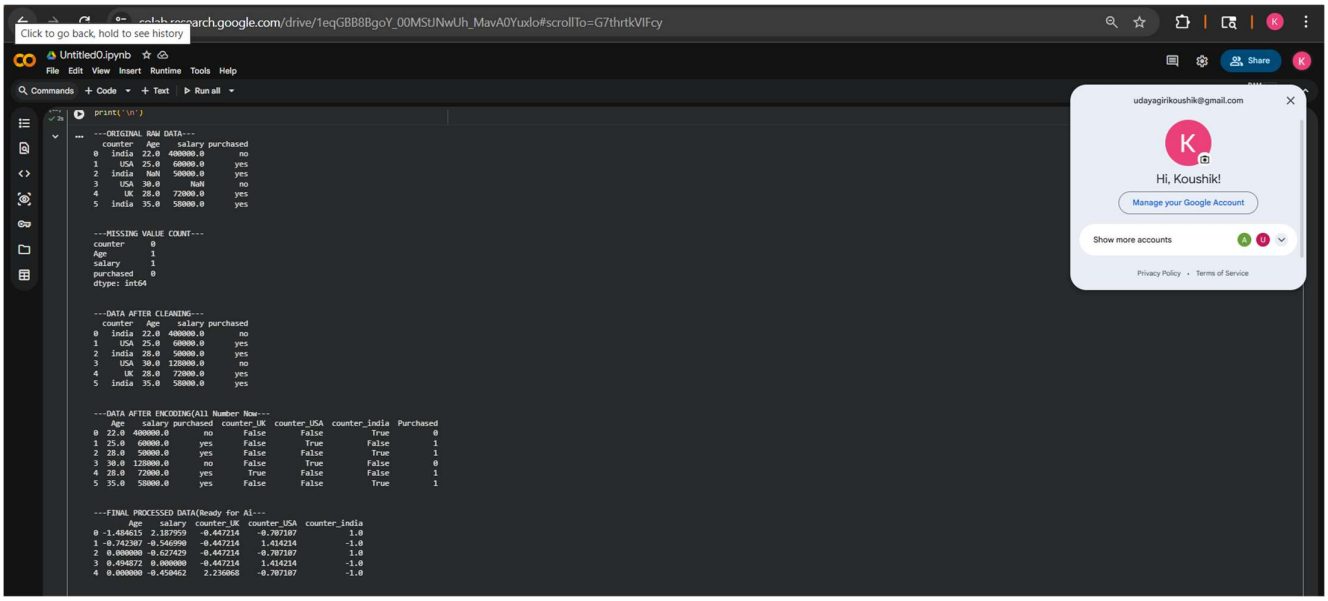
Aim:

To perform data exploration and preprocessing using Python.

PROGRAM:



Output:



Result:

The dataset was explored, cleaned, and pre processed successfully, making it ready for further analysis or model building.

IMPLEMENT REGULARIZED LINEAR REGRESSION

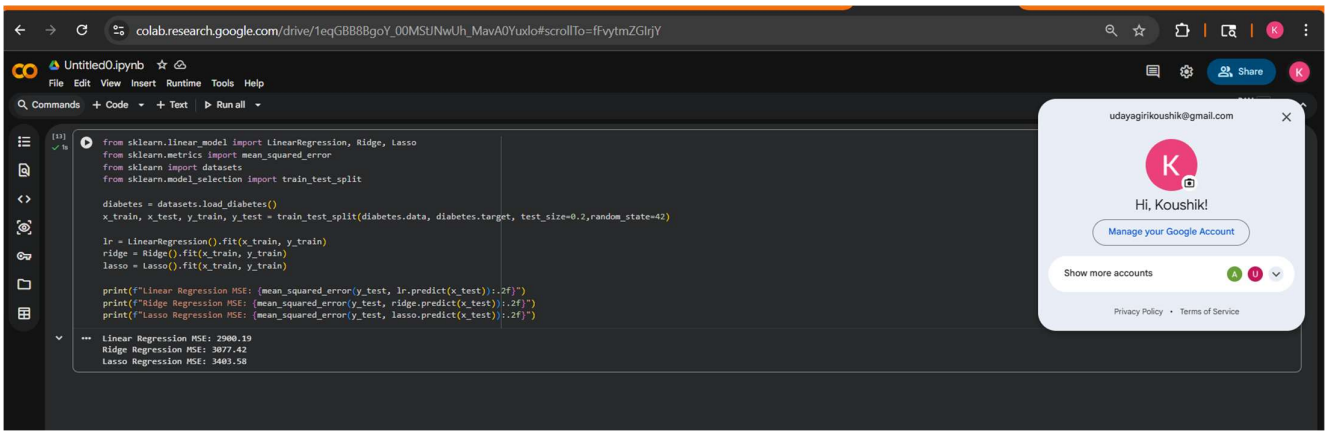
EXP-03

DATE-14-02-2026

Aim

To implement Regularized Linear Regression using Python.

PROGRAM WITH OUTPUT



Result

The regularized linear regression model was successfully implemented using Python. The model reduced overfitting and improved prediction performance by applying regularization techniques such as Ridge or Lasso regression.

IMPLEMENT NAVIE BAYES CLASSIFIER

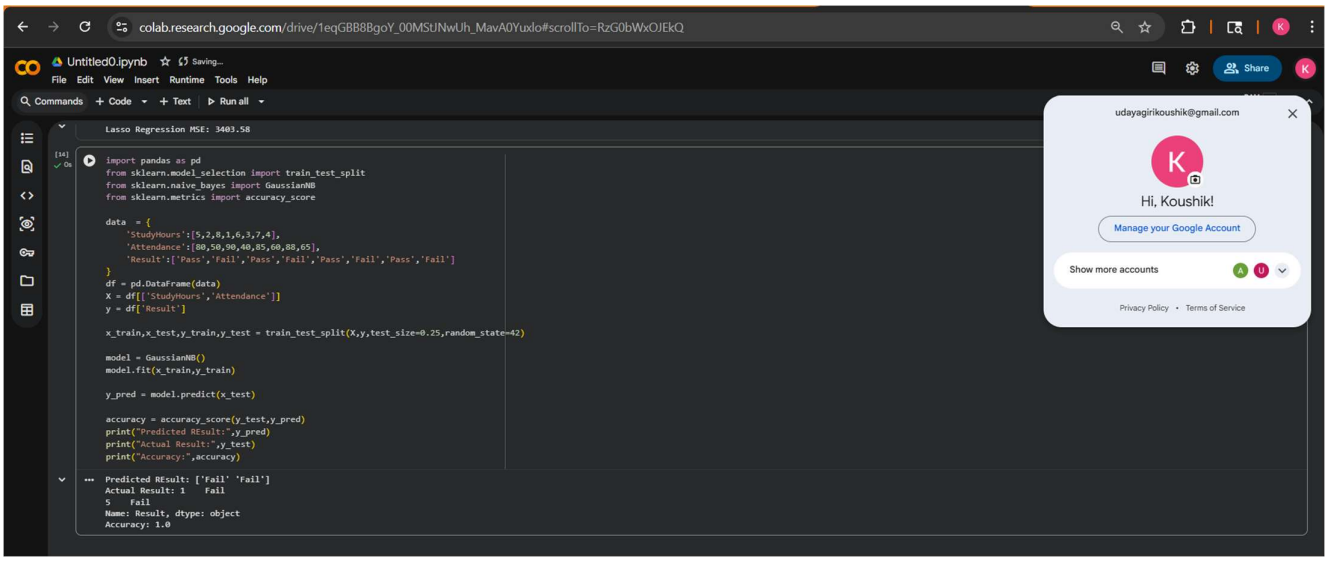
EXP-04

DATE-23-02-2026

Aim

To implement a Naive Bayes classifier using Python for classification of data.

PROGRAM WITH OUTPUT



Result

The Naive Bayes classifier was successfully implemented using Python. The model was trained and used to classify the data effectively, providing accurate predictions based on probability.

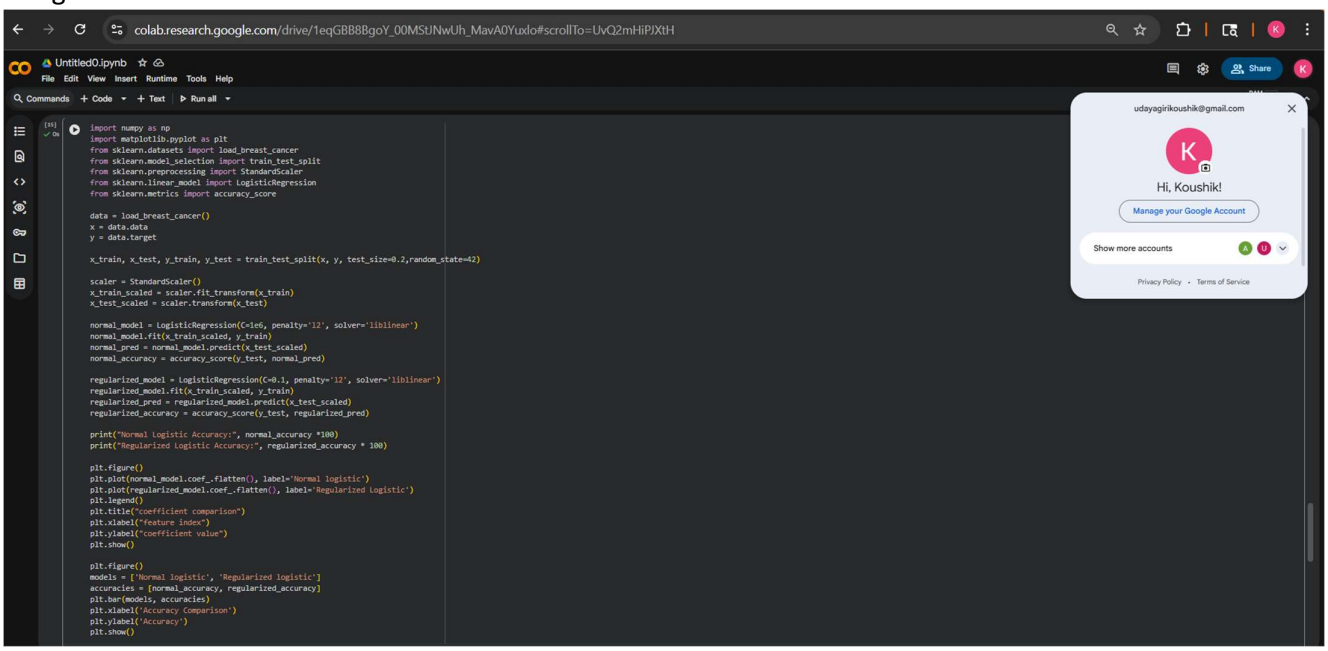
IMPLEMENT REGULARIZED LOGISTIC REGRESSION

EXP-05

DATE-02-03-2026 Aim:

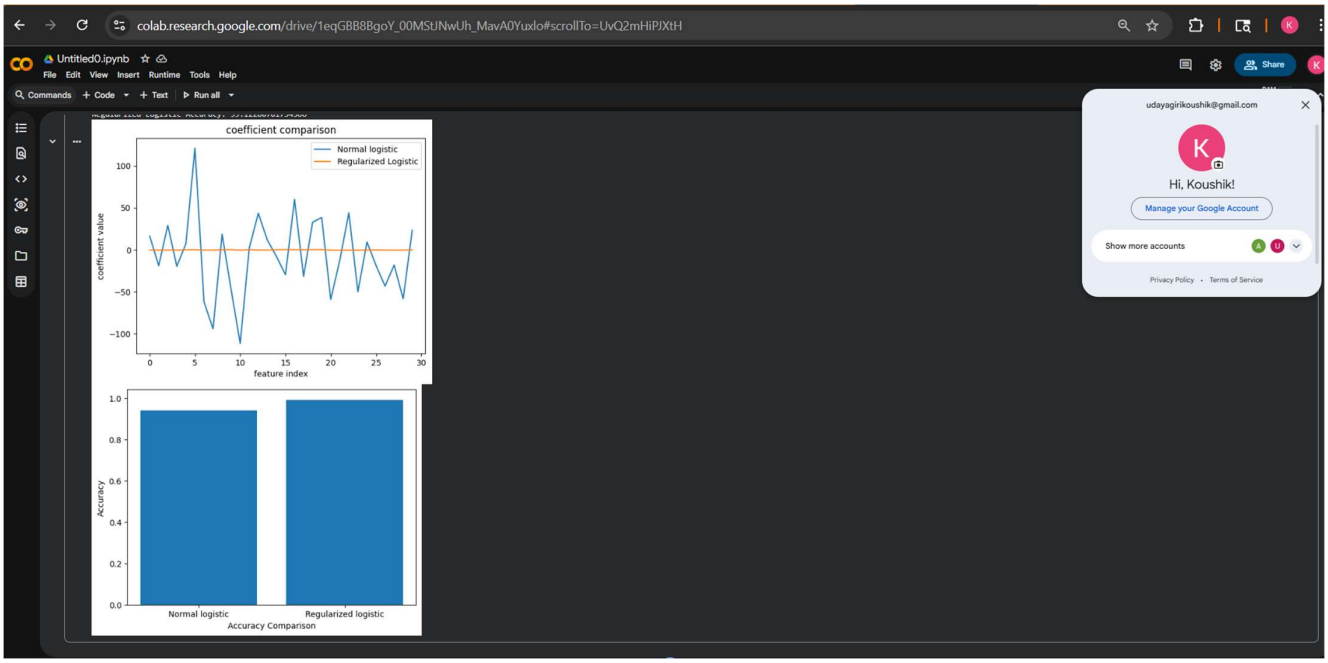
To implement Regularized Logistic Regression using Python.

Program:



```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

data = load_breast_cancer()
x = data.data
y = data.target

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2,random_state=42)

scaler = StandardScaler()
x_train_scaled = scaler.fit_transform(x_train)
x_test_scaled = scaler.transform(x_test)

normal_model = LogisticRegression(C=1e6, penalty='l2', solver='liblinear')
normal_model.fit(x_train_scaled, y_train)
normal_pred = normal_model.predict(x_test_scaled)
normal_accuracy = accuracy_score(y_test, normal_pred)

regularized_model = LogisticRegression(C=0.1, penalty='l2', solver='liblinear')
regularized_model.fit(x_train_scaled, y_train)
regularized_pred = regularized_model.predict(x_test_scaled)
regularized_accuracy = accuracy_score(y_test, regularized_pred)

print("Normal Logistic Accuracy:", normal_accuracy *100)
print("Regularized Logistic Accuracy:", regularized_accuracy * 100)

plt.figure()
plt.plot(normal_model.coef_.flatten(), label='Normal logistic')
plt.plot(regularized_model.coef_.flatten(), label='Regularized logistic')
plt.legend()
plt.title("coefficient comparison")
plt.xlabel("feature index")
plt.ylabel("coefficient value")
plt.show()

plt.figure()
models = ['Normal logistic', 'Regularized logistic']
accuracies = [normal_accuracy, regularized_accuracy]
plt.bar(models, accuracies)
plt.xlabel('Accuracy Comparison')
plt.ylabel('Accuracy')
plt.show()
```

OUTPUT-



Result

The regularized logistic regression model was successfully implemented using Python. Regularization helped reduce overfitting and improved the model's classification performance and generalization on the dataset.

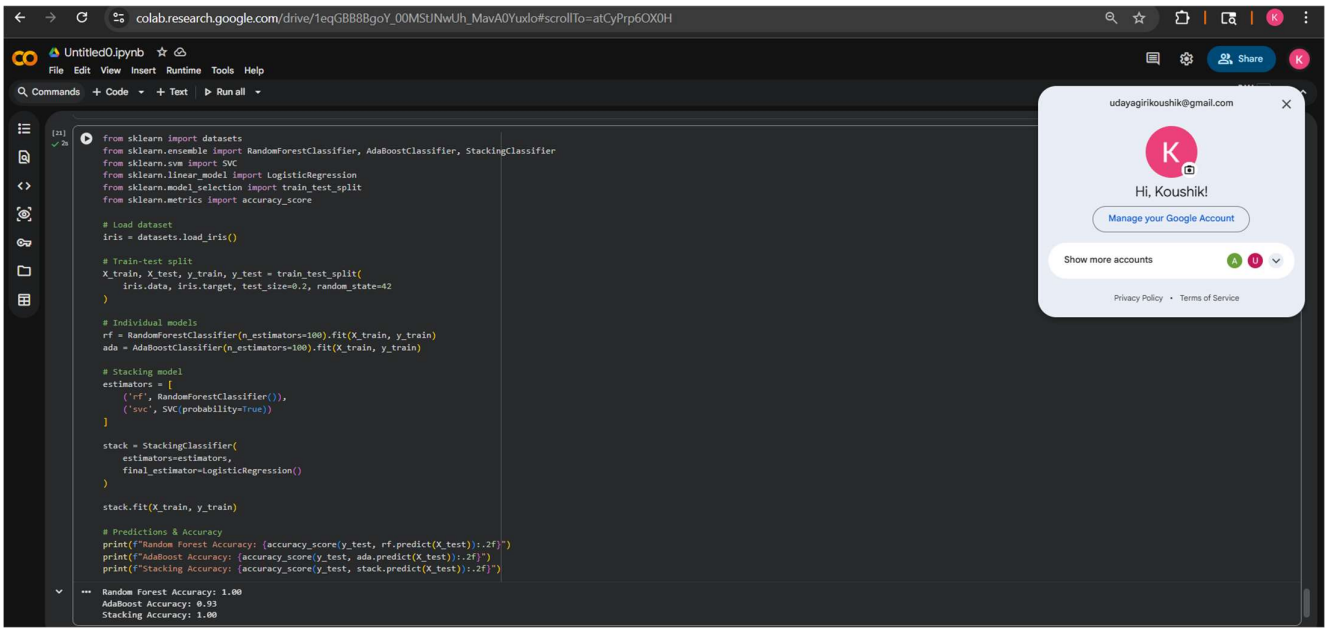
BUILD MODULES USING DIFFERENT ENSEMBLING TECHNIQUES

EXP-06

DATE-09-03-2026 Aim:

To build models using different ensemble techniques in Python.

PROGRAM WITH OUTPUT



Result

The ensemble models were successfully built using techniques such as Bagging, Boosting, and Random Forest. These methods improved prediction accuracy and reduced overfitting compared to individual models.

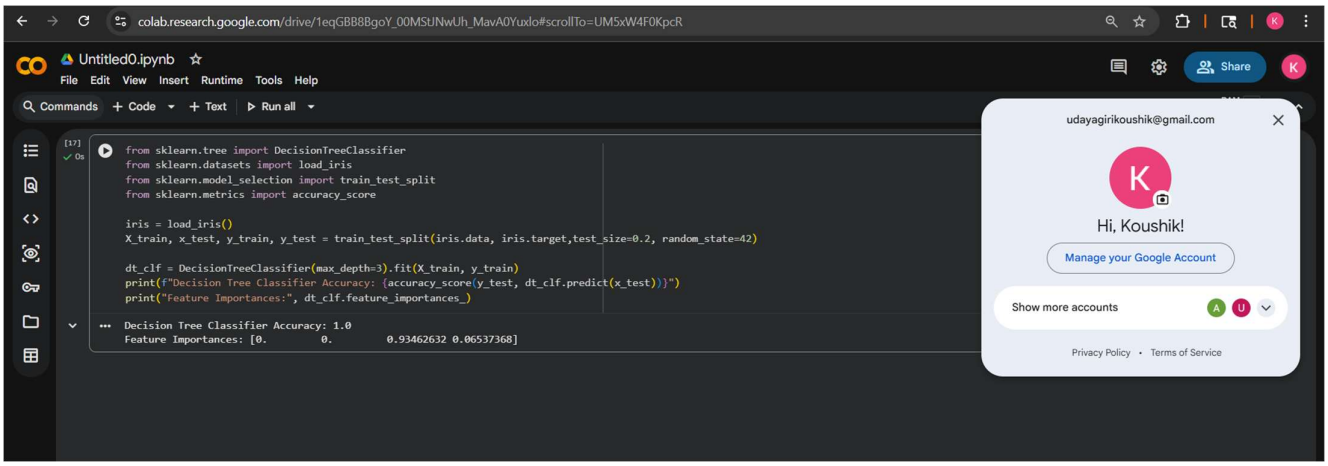
BUILD MODELS USING DECESION TREES

EXP-7

DATE-09-03-2026 Aim:

To build a classification model using Decision Trees in Python.

PROGRAM WITH OUTPUT



Result:

The Decision Tree model was successfully implemented using Python. The model learned decision rules from the dataset and provided accurate predictions for classification tasks.

BUILD THE MODELS USING DIFFERENT KERNAL FUNCTIONS

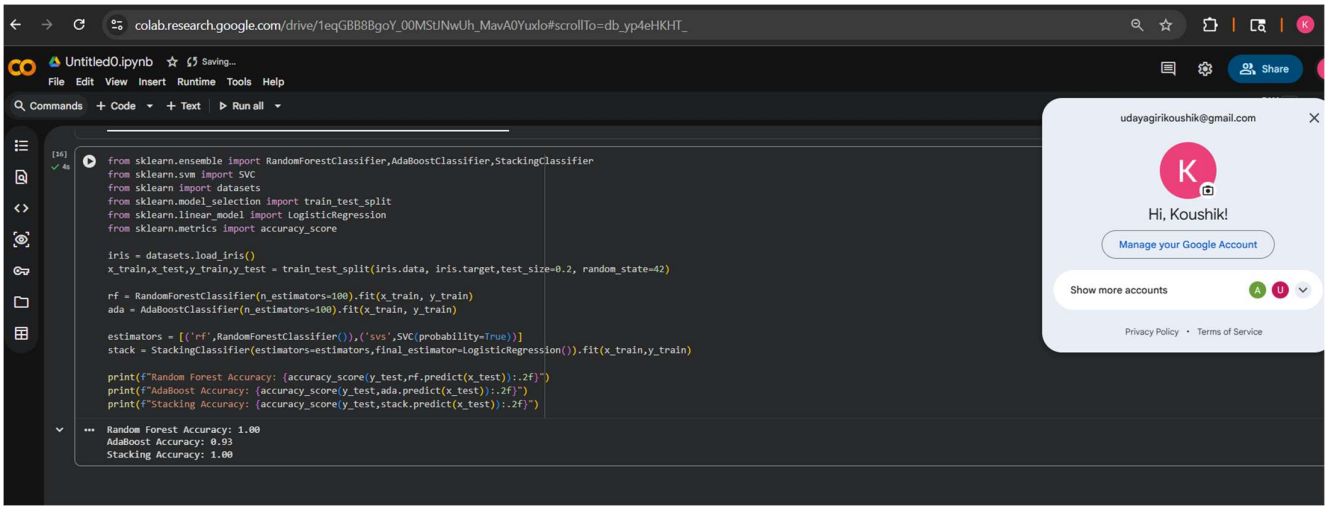
EXP-08

DATE-16-03-2026

Aim

To build models using different kernel functions in Python.

PROGRAM WITH OUTPUT



Result

The model was successfully implemented using kernel-based methods (such as SVM). Different kernel functions like linear, polynomial, and RBF were used, and they improved classification performance based on the dataset structure.

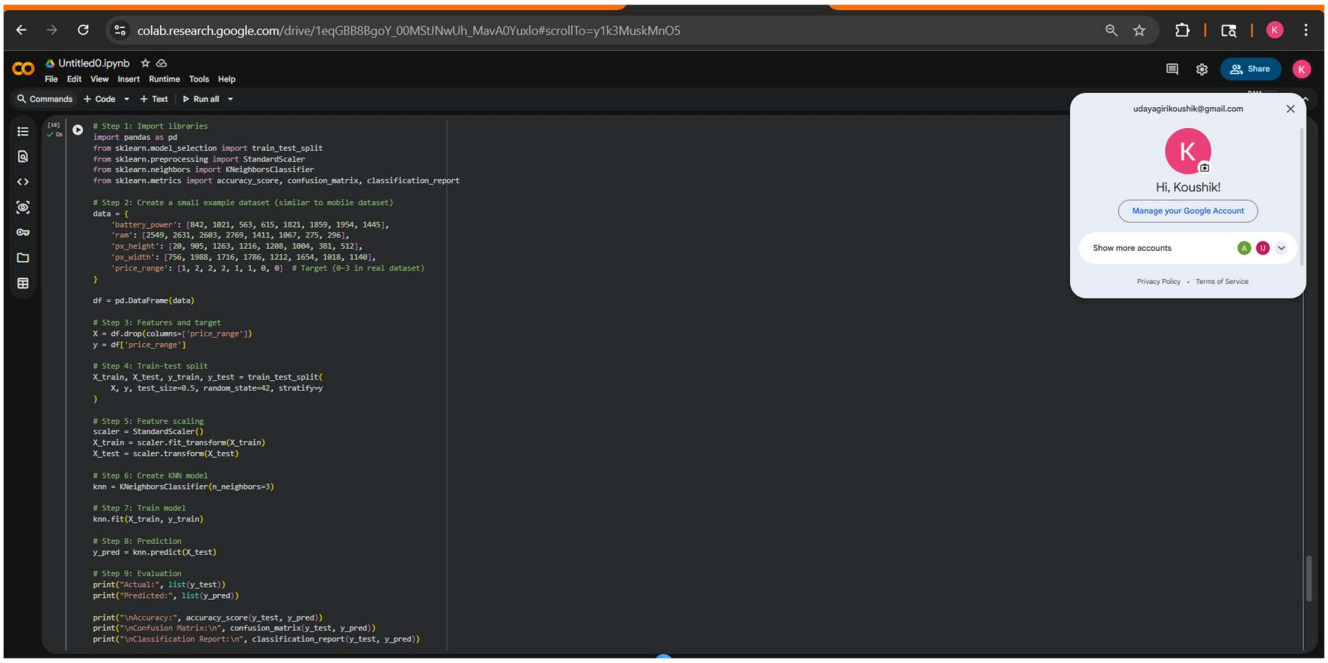
IMPLEMENT K-NEAREST NEIGHBOURS (KNN)CLASSIFICATION

EXP-09 DATE-30-3-

2026 Aim:

To implement the K-Nearest Neighbours (KNN) classification algorithm using Python.

PROGRAM WITH OUTPUT



Result

The KNN classification model was successfully implemented using Python. The model classified data based on the nearest neighbors and produced effective and accurate predictions for the given dataset.

K-MEAN CLUSTERING WITH PCA AND ELBOW METHOD

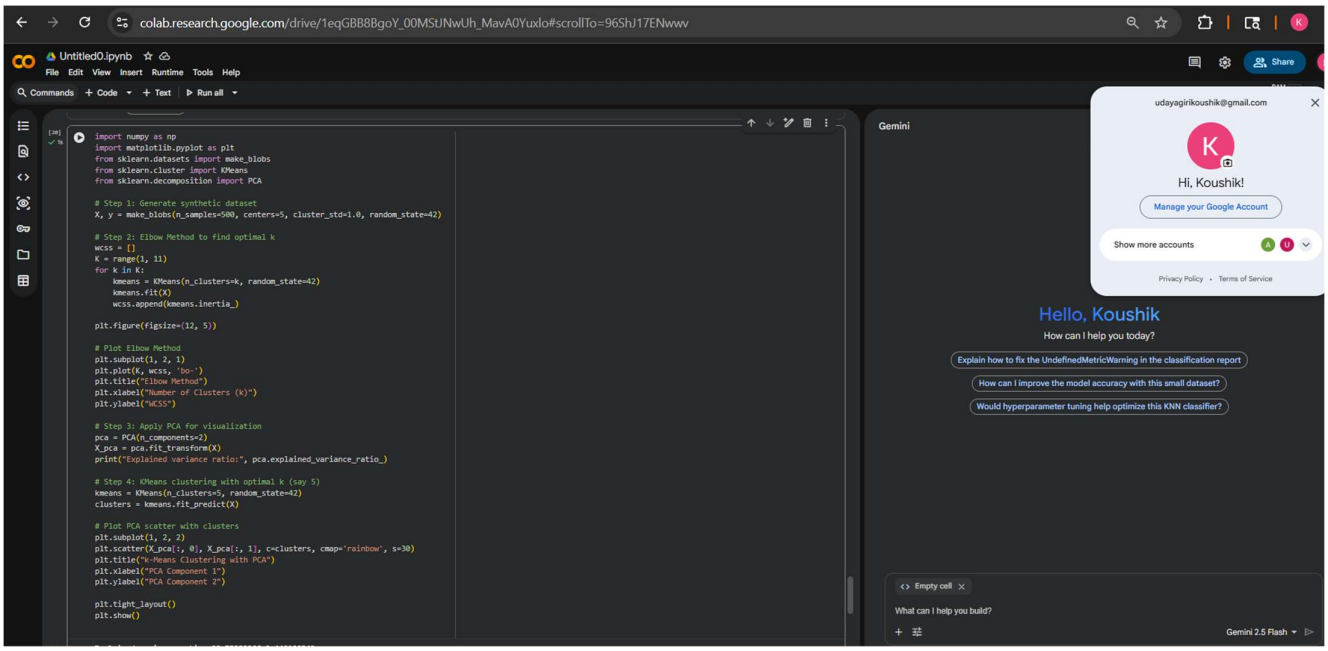
EXP-10

DATE-30-03-2026

Aim

To perform K-Means clustering using Python along with PCA and the Elbow method.

Program:



OUTPUT

Result

K-Means clustering was successfully implemented. The optimal number of clusters was identified using the Elbow method, and PCA was used to reduce dimensionality for better visualization. The data was grouped into meaningful clusters effectively.